

Multiple Polynomial evaluation reduction

Pushpendra Pal

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1 Need

Take the case of counting triangles in a graph; there we had a function in the MLE version.

$$\tilde{g}(\text{Graph}) = \frac{1}{6} \sum_i \sum_j \sum_k \tilde{f}(i, j) \cdot \tilde{f}(j, k) \cdot \tilde{f}(k, i) \quad (1)$$

Here, we need to evaluate $\tilde{f}$ at three points each time. How to do better?

2 Idea

- let us begin by supposing that $\tilde{W}$ is a MLE polynomial over F with $\log n$ variables, and the verifier wishes to evaluate $\tilde{W}$ at just two points, say $b, c \in F^{\log n}$
- A simple one round IP with communication cost $O(\log n)$ that reduces the evaluation of $\tilde{W}(b)$ and $\tilde{W}(c)$ to the evaluation of $\tilde{W}(r)$ for a single point $r \in F^{\log n}$.
- What this means is that the protocol will force the prover P to send claimed values v_0 and v_1 for $\tilde{W}(b)$ and $\tilde{W}(c)$, as well as claimed values for many other points chosen by the verifier V in a specific manner.
- V will then pick r at random from those points, and it will be safe for V to believe that $v_0 = \tilde{W}(b)$ and $v_1 = \tilde{W}(c)$ as long as P 's claim about $\tilde{W}(r)$ is valid.
- In other words, the protocol will ensure that if either $v_0 \neq \tilde{W}(b)$ or $v_1 \neq \tilde{W}(c)$, then with high probability over the V 's choice of r , it will also be the case that the prover makes a false claim as to the value of $\tilde{W}(r)$.

3 Protocol

- $l : F \rightarrow F^{\log n}$ be a line passing through b and c such that $l(0) = b$ and $l(1) = c$.
- P sends a uni-variate polynomial q ($\deg \leq \log n$) claimed to be $\tilde{W} \circ l$, restricting $\tilde{W}$ to l .
- V interprets $q(i) = v_i$, where $v_0 = \tilde{W}(b)$ and $v_1 = \tilde{W}(c)$.

4 Example

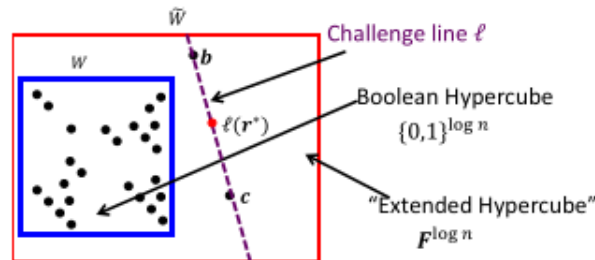


Figure 1:

- Suppose that $\log n = 2$, $b = (2, 4)$, $c = (3, 2)$, and $\tilde{W}(x_1, x_2) = 3x_1x_2 + 2x_2$.

- Line l is unique with $l(0) = b$ and $l(1) = c$ with paramatrix form $t \rightarrow (t + 2, 4 - 2t)$.
- The restriction of $\widetilde{W}$ to $l = \widetilde{W} \circ l = 3(t + 2)(4 - 2t) + 2(4 - 2t) = -6t^2 - 4t + 32$.
- **If P sends a degree-2 uni-variate polynomial q claimed to equal $\widetilde{W} \circ l$, the verifier will interpret $q(0)$ and $q(1)$ as claims about $\widetilde{W}(b)$ and $\widetilde{W}(c)$ respectively.**
- The verifier V will then pick a random $r^* \in F$, set $r = l(r^*)$ and interpret $q(r^*)$ as the claimed value of $\widetilde{W}(r)$.
- Observe that $l(r) = (r + 2, 4 - 2r)$ is a random point on the line l .